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COURSE CODE: SBT-DF202
LAB NAME: DIGITAL FORENSICS CASE HANDLING, AUTOPSY AND SLEUTH KIT ANALYSIS
DATE: 29/08/2026
INSTRUCTOR: AMINU IDRIS

Laboratory instructions

Before beginning this lab, review the **Module 1 and Module 2 lessons and study materials** covering digital investigation procedures, evidence handling, chain of custody, forensic imaging, Autopsy and The Sleuth Kit.

Download the authorised training forensic image [Ch01InChap01.dd](#) and create a dedicated forensic working directory before beginning your analysis. If the direct download does not start automatically, use the [browser download page](#) instead. The source lab specifically provides this evidence image for the practical.

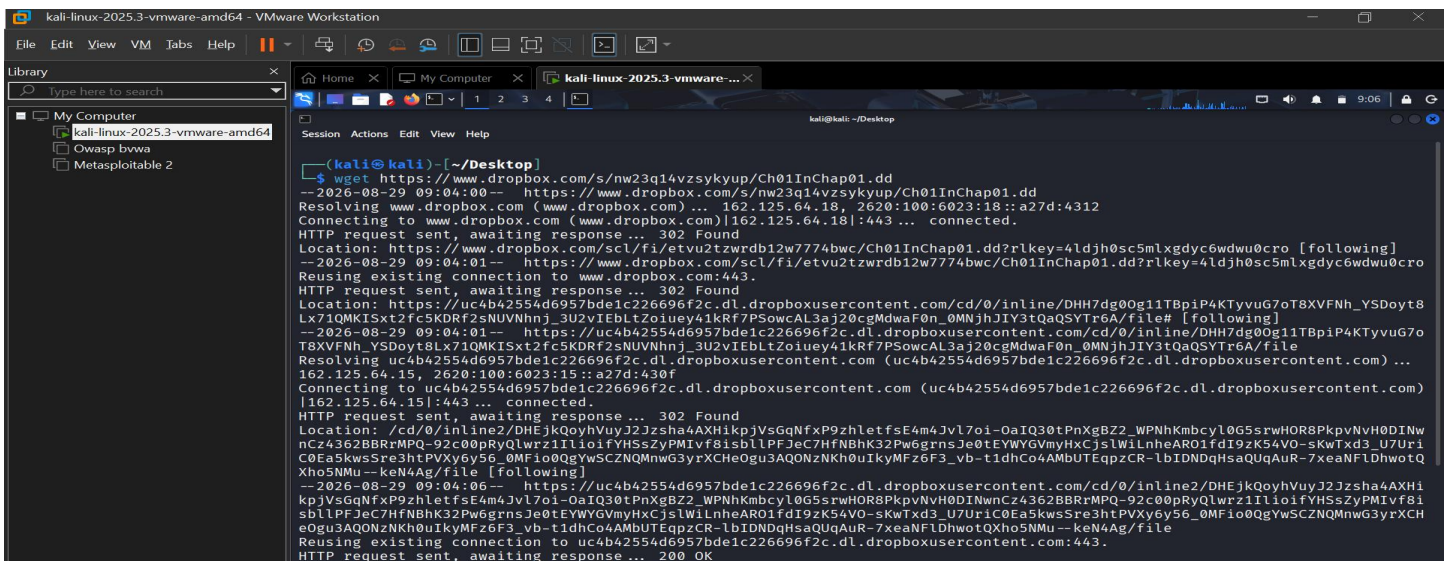
After downloading the evidence image, record the evidence source, file size, storage location, MD5 hash and SHA-256 hash before conducting any forensic analysis. Preserve the original image and perform recovery and examination activities in separate working folders.

Part A — Autopsy Analysis

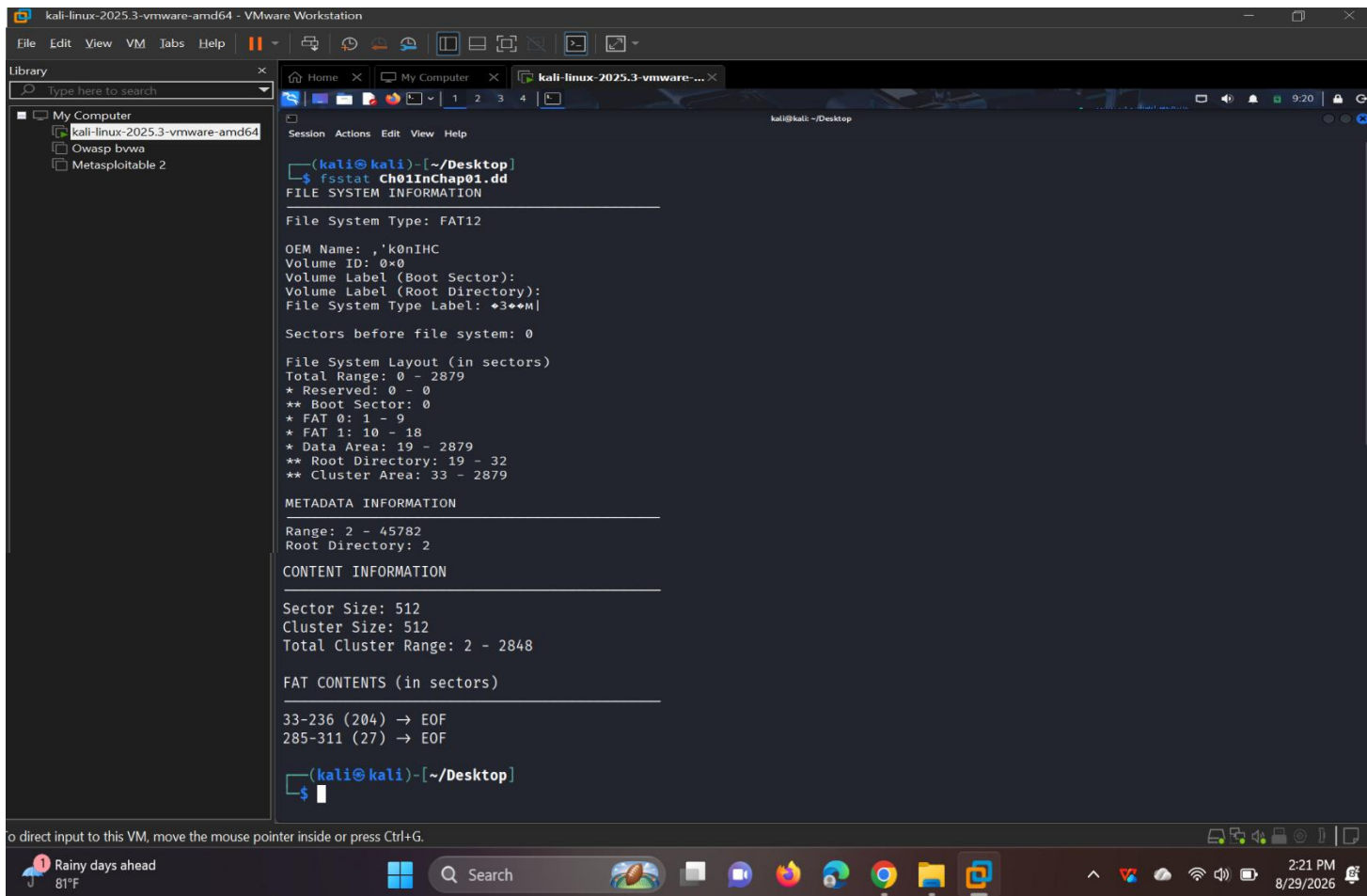
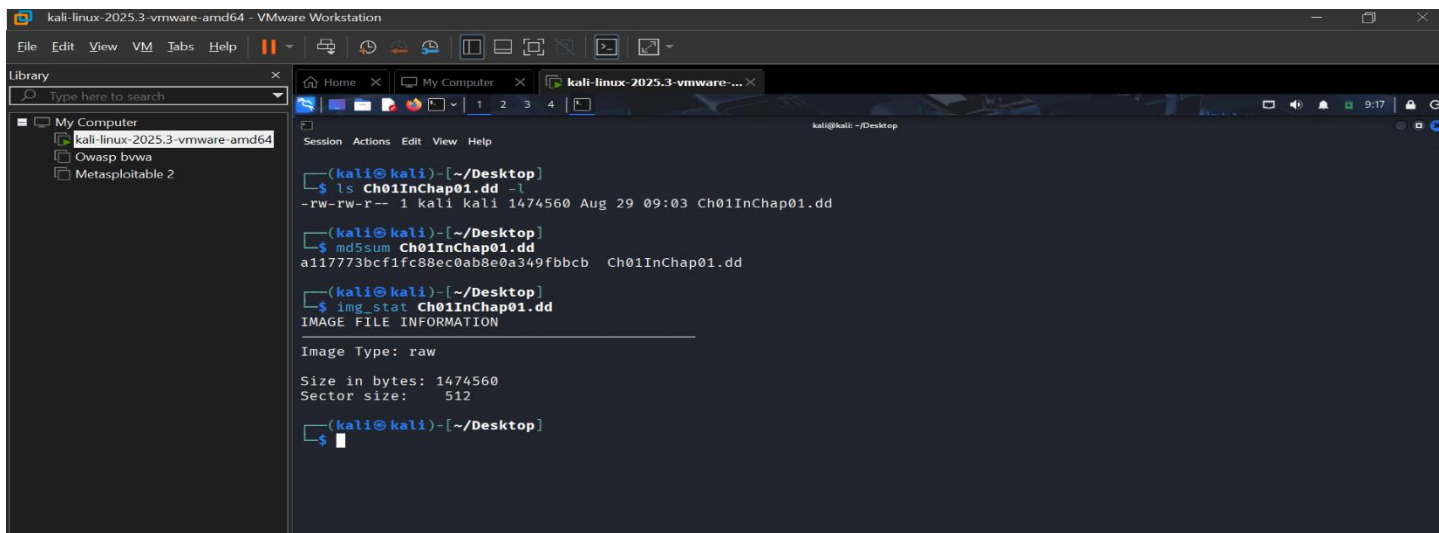
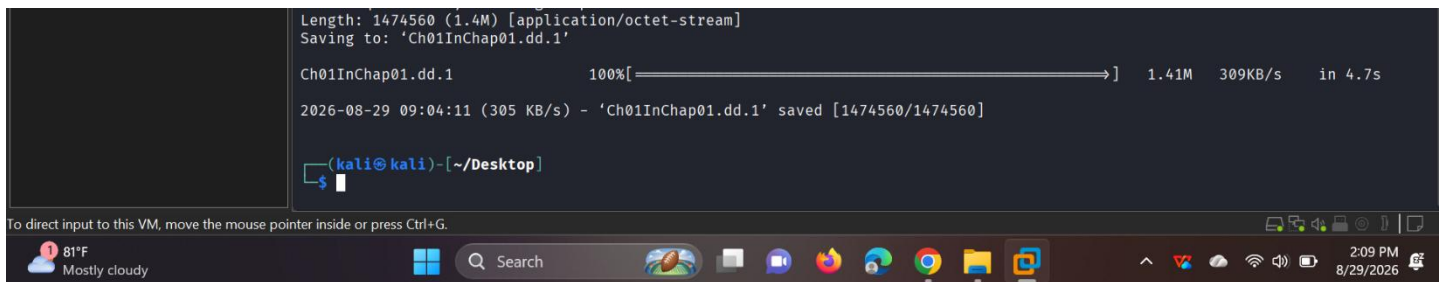
Create a forensic case in Autopsy and add **Ch01InChap01.dd** as the evidence source. Identify deleted files, recover relevant files, perform keyword searches, tag relevant evidence and generate an Autopsy report. Record the case name, evidence source, deleted-file findings, recovered files, keyword-search results and reporting evidence.

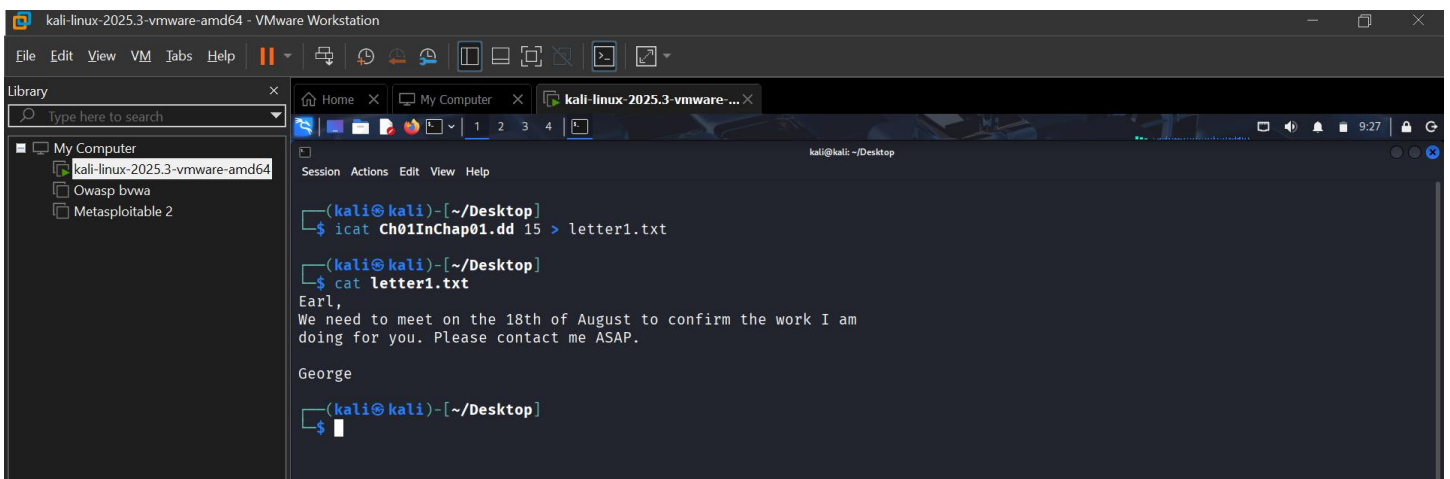
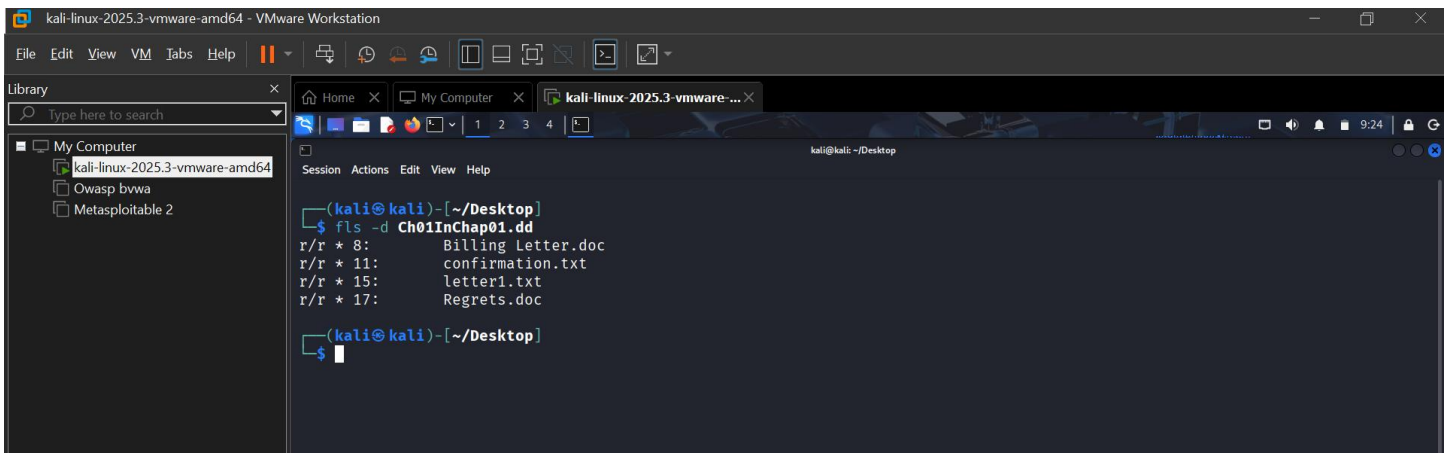
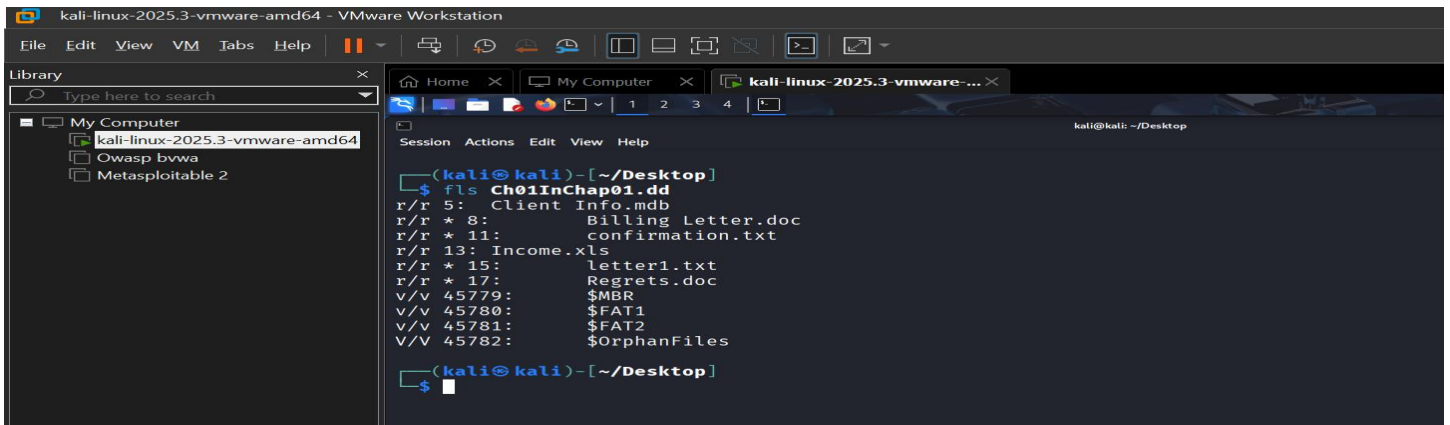
ANSWER:

File Name: Ch01InChap01.dd
File Size in bytes: 1474560
Sector Size: 512
MD5 hash: a117773bcf1fc88ec0ab8e0a349fbbcb Ch01InChap01.dd
SHA-256 hash: Image Type: raw



```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
kali-linux-2025.3-vmware-amd64
Owasp bwaa
Metasploitable 2
Session Actions Edit View Help
(kali@kali) ~ - /Desktop
$ wget https://www.dropbox.com/s/nw23q14vzsykyup/Ch01InChap01.dd
--2026-08-29 09:04:00-- https://www.dropbox.com/s/nw23q14vzsykyup/Ch01InChap01.dd
Resolving www.dropbox.com (www.dropbox.com)... 162.125.64.18, 2620:100:6023:18::a27d:4312
Connecting to www.dropbox.com (www.dropbox.com)|162.125.64.18|:443 ... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://www.dropbox.com/scl/fi/etvu2tzwrd12w7774bwc/Ch01InChap01.dd?rlkey=4ldjh0sc5mlxgdy6wdwu0cro [following]
--2026-08-29 09:04:01-- https://www.dropbox.com/scl/fi/etvu2tzwrd12w7774bwc/Ch01InChap01.dd?rlkey=4ldjh0sc5mlxgdy6wdwu0cro
Reusing existing connection to www.dropbox.com:443.
HTTP request sent, awaiting response... 302 Found
Location: https://uc4b42554d6957bde1c226696f2c.dl.dropboxusercontent.com/cd/0/inline/DHH7dg00g11TBpiP4KTyvuG7oT8XVFNh_YSDoyt8Lx71QMISxt2fc5KDRF2sNUVNHnj_3U2vIEbltZoiuey41kRf7PSowcAL3aj20cgMdwaf0n_0MNjhJIY3tQaQSYTr6A/file# [following]
--2026-08-29 09:04:01-- https://uc4b42554d6957bde1c226696f2c.dl.dropboxusercontent.com/cd/0/inline/DHH7dg00g11TBpiP4KTyvuG7oT8XVFNh_YSDoyt8Lx71QMISxt2fc5KDRF2sNUVNHnj_3U2vIEbltZoiuey41kRf7PSowcAL3aj20cgMdwaf0n_0MNjhJIY3tQaQSYTr6A/file
Resolving uc4b42554d6957bde1c226696f2c.dl.dropboxusercontent.com (uc4b42554d6957bde1c226696f2c.dl.dropboxusercontent.com)...
162.125.64.15, 2620:100:6023:15::a27d:430f
Connecting to uc4b42554d6957bde1c226696f2c.dl.dropboxusercontent.com (uc4b42554d6957bde1c226696f2c.dl.dropboxusercontent.com)|162.125.64.15|:443 ... connected.
HTTP request sent, awaiting response... 302 Found
Location: /cd/0/inline2/DHEjkQoyhVuyJ2Jzsha4AXHikpJVsGqNfxP9zhletfsE4m4JvL7oi-OaIQ30tPnXgBZ2_WPNhKmbcyL0G5srwHOR8PkpVNVH0DINwnCz4362B8RrMPQ-92c00pRyQLwrz1IlioifYHSsZyPMivf81sb1LPFJec7HFNhK32Pw6grnsJe0tEYWYGVmyHxCjSlWlLnheAR01fdI9zK54VO-sKwTxd3_U7UurIC0Ea5kwsSre3htPVXygy56_0MFio0QgYwSCZNQmNwG3yrXCHeogU3AQONZKk0uikyMF26F3_vb-t1dhCo4AMbUTEqpzCR-lbIDNDQhSaQuAuR-7xeaNF1DhwotQXho5NMu--keN4Ag/file [following]
--2026-08-29 09:04:06-- https://uc4b42554d6957bde1c226696f2c.dl.dropboxusercontent.com/cd/0/inline2/DHEjkQoyhVuyJ2Jzsha4AXHikpJVsGqNfxP9zhletfsE4m4JvL7oi-OaIQ30tPnXgBZ2_WPNhKmbcyL0G5srwHOR8PkpVNVH0DINwnCz4362B8RrMPQ-92c00pRyQLwrz1IlioifYHSsZyPMivf81sb1LPFJec7HFNhK32Pw6grnsJe0tEYWYGVmyHxCjSlWlLnheAR01fdI9zK54VO-sKwTxd3_U7UurIC0Ea5kwsSre3htPVXygy56_0MFio0QgYwSCZNQmNwG3yrXCHeogU3AQONZKk0uikyMF26F3_vb-t1dhCo4AMbUTEqpzCR-lbIDNDQhSaQuAuR-7xeaNF1DhwotQXho5NMu--keN4Ag/file
Reusing existing connection to uc4b42554d6957bde1c226696f2c.dl.dropboxusercontent.com:443.
HTTP request sent, awaiting response... 200 OK
```



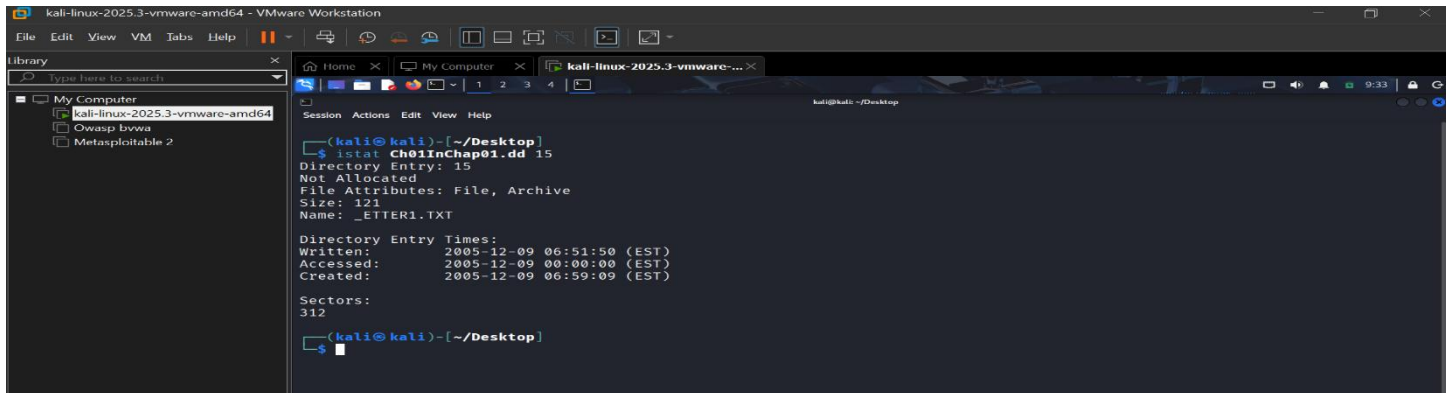


Part B — Sleuth Kit Analysis

Examine the forensic image using `img_stat`, `mmls`, `fsstat` and `fls`. Identify deleted files and locate **INCOME.XLS**. Determine its metadata address, examine its metadata and data units using `istat`, recover it using `icat` and `blkcat` where applicable, perform additional recovery using `tsk_recover`, and compare the hashes of the recovered copies.

Document every significant command, output, observation, error and corrective action. Screenshots must clearly support the work claimed in your report.

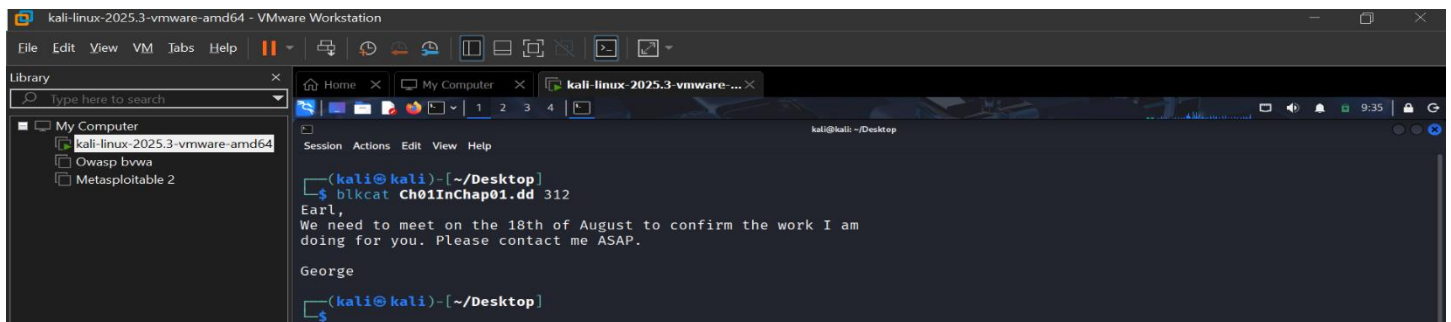
ANSWER:



```
kali@kali: ~/Desktop
$ lsstat Ch01InChap01.dd 15
Directory Entry: 15
Not Allocated
File Attributes: File, Archive
Size: 121
Name: _ETTER1.TXT

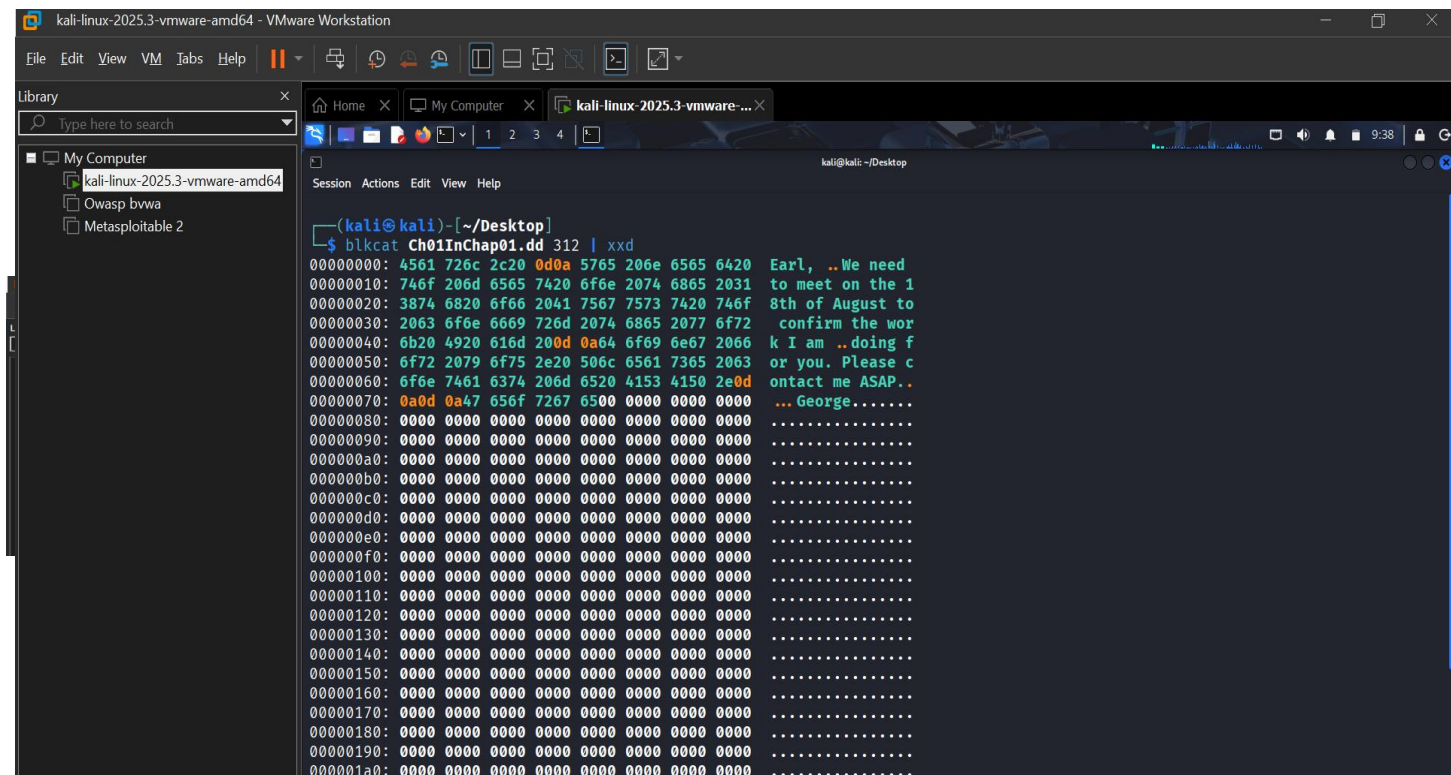
Directory Entry Times:
Written: 2005-12-09 06:51:50 (EST)
Accessed: 2005-12-09 00:00:00 (EST)
Created: 2005-12-09 06:59:09 (EST)

Sectors:
312
```



```
kali@kali: ~/Desktop
$ blkcat Ch01InChap01.dd 312
Earl,
We need to meet on the 18th of August to confirm the work I am
doing for you. Please contact me ASAP.

George
```



```
kali@kali: ~/Desktop
$ blkcat Ch01InChap01.dd 312 | xxd
00000000: 4561 726c 2c20 0d0a 5765 206e 6565 6420  Earl, ..We need
00000010: 746f 206d 6565 7420 6f6e 2074 6865 2031  to meet on the 1
00000020: 3874 6820 6f66 2041 7567 7573 7420 746f  8th of August to
00000030: 2063 6f6e 6669 726d 2074 6865 2077 6f72  confirm the wor
00000040: 6b20 4920 616d 200d 0a64 6f69 6e67 2066  k I am ..doing f
00000050: 6f72 2079 6f75 2e20 506c 6561 7365 2063  or you. Please c
00000060: 6f6e 7461 6374 206d 6520 4153 4150 2e0d  ontact me ASAP..
00000070: 0a0d 0a47 656f 7267 6500 0000 0000 0000  ... George.....
00000080: 0000 0000 0000 0000 0000 0000 0000 0000  .....
00000090: 0000 0000 0000 0000 0000 0000 0000 0000  .....
000000a0: 0000 0000 0000 0000 0000 0000 0000 0000  .....
000000b0: 0000 0000 0000 0000 0000 0000 0000 0000  .....
000000c0: 0000 0000 0000 0000 0000 0000 0000 0000  .....
000000d0: 0000 0000 0000 0000 0000 0000 0000 0000  .....
000000e0: 0000 0000 0000 0000 0000 0000 0000 0000  .....
000000f0: 0000 0000 0000 0000 0000 0000 0000 0000  .....
00000100: 0000 0000 0000 0000 0000 0000 0000 0000  .....
00000110: 0000 0000 0000 0000 0000 0000 0000 0000  .....
00000120: 0000 0000 0000 0000 0000 0000 0000 0000  .....
00000130: 0000 0000 0000 0000 0000 0000 0000 0000  .....
00000140: 0000 0000 0000 0000 0000 0000 0000 0000  .....
00000150: 0000 0000 0000 0000 0000 0000 0000 0000  .....
00000160: 0000 0000 0000 0000 0000 0000 0000 0000  .....
00000170: 0000 0000 0000 0000 0000 0000 0000 0000  .....
00000180: 0000 0000 0000 0000 0000 0000 0000 0000  .....
00000190: 0000 0000 0000 0000 0000 0000 0000 0000  .....
000001a0: 0000 0000 0000 0000 0000 0000 0000 0000  .....
```

